

Language Definition

1 Abstract Syntax

1.1 Values

$$v ::= \begin{array}{l} n \\ | \quad b \\ | \quad C(v_1, \dots, v_n) \\ | \quad f_1 : v_1, \dots, f_m : v_m \end{array}$$

where $n \in \mathbb{Z}$, $b \in \{\mathbf{true}, \mathbf{false}\}$, and C is a constructor name.

1.2 Types

$$T ::= \begin{array}{l} \mathbf{int} \\ | \quad \mathbf{bool} \\ | \quad V \\ | \quad R \end{array}$$

Variant types are labeled sums:

$$V ::= \langle C_1 : \overline{T}_1, \dots, C_n : \overline{T}_n \rangle$$

where each $\overline{T}_i$ denotes a (possibly empty) sequence of types and C_i are constructor names.

Record types are labeled products:

$$R ::= \{f_1 : T_1, \dots, f_m : T_m\}$$

where each f_i is a field name and T_i is its corresponding type.

1.3 Expressions

$$e ::= \begin{array}{l} v \\ | \quad e_1 \mathbf{op} e_2 \\ | \quad ! e \\ | \quad x \end{array}$$

where $\mathbf{op} \in \{+, -, *, /, \wedge, \vee, ==, <, >\}$ and x is a variable name.

1.4 Statements

$$\begin{array}{l}
 s ::= \text{var } x : T \\
 \quad | \quad x := e \\
 \quad | \quad \text{if } e \{s^*\} \\
 \quad | \quad \text{if } e \{s^*\} \text{ else } \{s^*\} \\
 \quad | \quad \text{print } e \\
 \quad | \quad \text{variant } V = C_1(\overline{T_1}) \mid \cdots \mid C_n(\overline{T_n}) \\
 \quad | \quad \text{record } R = \{f_1 : T_1, \dots, f_m : T_m\}
 \end{array}$$

2 Typing Rules

$$\Gamma \vdash e : T$$

2.1 Base Types

$$\frac{}{\Gamma \vdash n : \text{int}} \text{T-INT} \qquad \frac{}{\Gamma \vdash \text{true} : \text{bool}} \text{T-BOOL-TRUE} \qquad \frac{}{\Gamma \vdash \text{false} : \text{bool}} \text{T-BOOL-FALSE}$$

2.2 Variables

$$\frac{x : T \in \Gamma}{\Gamma \vdash x : T} \text{T-VAR-USE}$$

2.3 Binary Operators

$$\begin{array}{cc}
 \frac{\Gamma \vdash e_1 : \text{int} \quad \Gamma \vdash e_2 : \text{int}}{\Gamma \vdash e_1 + e_2 : \text{int}} \text{T-ARITHMETIC} & \frac{\Gamma \vdash e_1 : \text{bool} \quad \Gamma \vdash e_2 : \text{bool}}{\Gamma \vdash e_1 \wedge e_2 : \text{bool}} \text{T-LOGICAL} \\
 \\
 \frac{\Gamma \vdash e_1 : \text{int} \quad \Gamma \vdash e_2 : \text{int}}{\Gamma \vdash e_1 < e_2 : \text{bool}} \text{T-COMPARISON} & \frac{\Gamma \vdash e_1 : T \quad \Gamma \vdash e_2 : T}{\Gamma \vdash e_1 == e_2 : \text{bool}} \text{T-EQUALITY}
 \end{array}$$

2.4 Unary Operators

$$\frac{\Gamma \vdash e : \text{bool}}{\Gamma \vdash \text{not } e : \text{bool}} \text{T-NOT}$$

2.5 User-Defined Types

$$\frac{\Gamma \vdash e_1 : T_1, \dots, e_n : T_n \quad C : (T_1, \dots, T_n) \in \langle C_1 : \overline{T_1}, \dots, C_n : \overline{T_n} \rangle = V}{\Gamma \vdash C(e_1, \dots, e_n) : V} \text{T-VARIANT-CONSTRUCTOR}$$

$$\frac{\Gamma \vdash e_1 : T_1, \dots, e_m : T_m \quad R = \{f_1 : T_1, \dots, f_m : T_m\}}{\Gamma \vdash \{f_1 : e_1, \dots, f_m : e_m\} : R} \text{T-RECORD-VALUE}$$

3 Statements

$$\frac{\Gamma \vdash e : T \quad x : T \in \Gamma}{\Gamma \vdash x := e \text{ ok}} \text{ T-ASSIGN}$$

$$\frac{\Gamma \vdash e : \text{bool} \quad \Gamma \vdash s^* \text{ ok}}{\Gamma \vdash \text{if } e \{s^*\} \text{ ok}} \text{ T-IF}$$

$$\frac{\Gamma \vdash e : \text{bool} \quad \Gamma \vdash s_1^* \text{ ok} \quad \Gamma \vdash s_2^* \text{ ok}}{\Gamma \vdash \text{if } e \{s_1^*\} \text{ else } \{s_2^*\} \text{ ok}} \text{ T-IF-ELSE}$$

$$\frac{\Gamma \vdash e : T}{\Gamma \vdash \text{print } e \text{ ok}} \text{ T-PRINT}$$

4 Operational Semantics

The operational semantics are given as a big-step evaluation relation:

$$\langle e, \sigma \rangle \rightarrow v$$

4.1 Base Values

$$\frac{}{\langle n, \sigma \rangle \rightarrow n} \text{ E-INT} \qquad \frac{}{\langle \mathbf{b}, \sigma \rangle \rightarrow \mathbf{b}} \text{ E-BOOL}$$

4.2 Variables

$$\frac{\sigma(x) = v}{\langle x, \sigma \rangle \rightarrow v} \text{ E-VAR}$$

4.3 Operations

$$\frac{\langle e_1, \sigma \rangle \rightarrow n_1 \quad \langle e_2, \sigma \rangle \rightarrow n_2}{\langle e_1 + e_2, \sigma \rangle \rightarrow n_1 + n_2} \text{ E-ARITHMETIC}$$

$$\frac{\langle e_1, \sigma \rangle \rightarrow b_1 \quad \langle e_2, \sigma \rangle \rightarrow b_2}{\langle e_1 \wedge e_2, \sigma \rangle \rightarrow b_1 \wedge b_2} \text{ E-LOGICAL}$$

$$\frac{\langle e, \sigma \rangle \rightarrow b}{\langle !e, \sigma \rangle \rightarrow \neg b} \text{ E-NOT}$$

$$\frac{\langle e_1, \sigma \rangle \rightarrow n_1 \quad \langle e_2, \sigma \rangle \rightarrow n_2}{\langle e_1 < e_2, \sigma \rangle \rightarrow (n_1 < n_2)} \text{ E-COMPARISON}$$

$$\frac{\langle e_1, \sigma \rangle \rightarrow v_1 \quad \langle e_2, \sigma \rangle \rightarrow v_2}{\langle e_1 == e_2, \sigma \rangle \rightarrow (v_1 == v_2)} \text{ E-EQUALITY}$$

4.4 Variants

$$\frac{\langle e_1, \sigma \rangle \rightarrow v_1 \quad \cdots \quad \langle e_n, \sigma \rangle \rightarrow v_n}{\langle C(e_1, \dots, e_n), \sigma \rangle \rightarrow C(v_1, \dots, v_n)} \text{ E-VARIANT-CONSTRUCTOR}$$

4.5 Records

$$\frac{\langle e_1, \sigma \rangle \rightarrow v_1 \quad \cdots \quad \langle e_n, \sigma \rangle \rightarrow v_n}{\langle \{f_1 : e_1, \dots, f_n : e_n\}, \sigma \rangle \rightarrow \{f_1 : v_1, \dots, f_n : v_n\}} \text{ E-RECORD-VALUE}$$

4.6 Statements

Statement execution is defined as:

$$\langle s, \sigma \rangle \rightarrow \sigma'$$

$$\frac{\langle e, \sigma \rangle \rightarrow v}{\langle x := e, \sigma \rangle \rightarrow \sigma[x \mapsto v]} \text{ E-ASSIGN}$$

$$\frac{\langle e, \sigma \rangle \rightarrow \mathbf{true} \quad \langle s^*, \sigma \rangle \rightarrow \sigma'}{\langle \mathbf{if } e \{s^*\}, \sigma \rangle \rightarrow \sigma'} \text{ E-IF-TRUE}$$

$$\frac{\langle e, \sigma \rangle \rightarrow \mathbf{false}}{\langle \mathbf{if } e \{s^*\}, \sigma \rangle \rightarrow \sigma} \text{ E-IF-FALSE}$$

$$\frac{\langle e, \sigma \rangle \rightarrow \mathbf{true} \quad \langle s_1^*, \sigma \rangle \rightarrow \sigma'}{\langle \mathbf{if } e \{s_1^*\} \mathbf{ else } \{s_2^*\}, \sigma \rangle \rightarrow \sigma'} \text{ E-IF-ELSE-TRUE}$$

$$\frac{\langle e, \sigma \rangle \rightarrow \mathbf{false} \quad \langle s_2^*, \sigma \rangle \rightarrow \sigma'}{\langle \mathbf{if } e \{s_1^*\} \mathbf{ else } \{s_2^*\}, \sigma \rangle \rightarrow \sigma'} \text{ E-IF-ELSE-FALSE}$$